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A lightweight framework for robust object detection in adverse weather based on dual-teacher feature alignment

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ABSTRACT

Object detection in adverse weather conditions (*e.g.,* rain, fog, snow) remains a critical challenge due to degraded image quality and semantic ambiguity. Mainstream approaches attempt to bridge this gap by aligning degraded images with their clear counterparts through two main paradigms: cascading enhancer-detector pipelines and multi-task learning frameworks that jointly optimize restoration and detection objectives. However, these methods often fail to reach their full potential due to the disparity between image restoration and detection tasks. Restoration networks prioritize pixel-level fidelity, while detection networks focus on high-level semantic accuracy, leading to suboptimal performance. In this work, we propose a Dual-Teacher Feature Alignment (DTFA) framework that rethinks the paradigm of "clear image alignment" for adverse weather detection. Instead of directly restoring pixel-level fidelity, we employ clear features as alignment to guide the training of the adverse weather detection student. Specifically, the Invariant Reconstruction Teacher (IRT) from a pre-trained reconstruction network provides weather-invariant priors, and the Semantic Prior Teacher (SPT) from a high-performance detection network offers task-aware semantic information. Two Adaptive Feature Bridging modules dynamically align multi-scale features between the two teachers and the adverse weather detection student, addressing task discrepancies through masked consistency constraints and efficiently enabling the adverse weather detection student to learn complementary information from both the IRT and SPT. During testing, only the adverse weather detection student remains. Therefore, no additional computational costs are incurred. Extensive experiments conducted in rainy, foggy, snowy, and mixed weather conditions demonstrate that our DTFA framework achieves state-of-the-art performance. The source code will be released at https://github.com/huruo1010/DTFA.

1. Introduction

Object detection is a fundamental task in computer vision, supporting critical applications such as autonomous vehicles, disaster response, and infrastructure monitoring. Although advancements [1–4] in deep learning have significantly enhanced detection accuracy in controlled experimental settings, their robustness declines markedly under adverse atmospheric conditions. Precipitation streaks, fog-induced haze, and snow occlusion introduce domain-specific noise and obstructions into image data, compromising model reliability and generalization.

While current detection frameworks [5–8] have achieved remarkable success in ideal lighting and visibility conditions, their performance significantly deteriorates when deployed in real-world environments, particularly those affected by rain, fog, or snow. A key reason is that these models are typically developed and evaluated on clean datasets, which do not reflect the complex degradations encountered in real-world conditions. Compounding this issue, acquiring a diverse and well-annotated real-world adverse weather dataset remains a significant bottleneck, as such conditions are challenging to systematically capture, costly to annotate, and often limited in both scale and variety. Consequently, addressing these challenges has become a critical research priority in multimedia computing, particularly for ensuring robust performance in uncontrolled outdoor settings.

The scarcity and high annotation costs of real-world adverse weather images present a significant bottleneck in developing robust detection models. To address this data acquisition challenge, a common strategy is to utilize synthetic data. Researchers introduce weather-specific degradations to clear images, generating synthetic adverse weather images with readily available clear counterparts. This approach facilitates the training of detection networks through image alignment or multi-task

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learning frameworks, which can be broadly categorized into enhancer-detector pipelines and multi-task learning paradigms, as summarized in Fig. 1.

It is a natural approach to employ enhancer-detector pipelines that utilize image restoration or enhancement techniques as a pre-processing step. A pre-trained, fixed restoration or enhancement network first processes the degraded images, after which the restored or enhanced images are transmitted to the detection network. Normally, they utilize the existing powerful restoration (or enhancement) networks for image deraining [9,10], defogging [11–15], and desnowing [16,17] to remove the degradation. The parameters of the detection network are updated using a detection loss. However, tasks such as image restoration and enhancement have distinct objectives and evaluation metrics when compared to object detection. Using image restoration or enhancement as a pre-processing step may not result in a more powerful detection model [18]. Recent advancements have investigated integrated solutions to bridge this gap. Some researchers [19–21] have proposed end-to-end enhancer-detector pipelines that jointly optimize restoration (or enhancement) and detection objectives. In these frameworks, the detection loss updates both the restoration (or enhancement) network and the detection network. IA-YOLO [19] introduces differentiable image processing to adaptively enhance images, thereby improving detection performance in foggy conditions. Similarly, GDIP-YOLO [20] utilizes a gating mechanism to dynamically integrate multiple pre-processing operators within the enhancement network. Along this line, MASFNet [21] proposes a multiscale adaptive sampling fusion network, which utilizes a Laplace pyramid-based feature enhancement module and a multiscale fusion structure to improve detection under various adverse weather conditions. Though these approaches demonstrate improved accuracy in adverse conditions, their architectural complexity from additional parameters inevitably compromises real-time detection performance.

The research community has subsequently shifted toward multi-task learning frameworks, aligning degraded images with their clear counterparts through a restoration decoder and a restoration loss. Several studies [22–25] have pioneered this framework by proposing shared-encoder architectures that include independent restoration and detection decoders. Specifically, the shared encoder is optimized using both the restoration loss from the restoration decoder and the detection loss from the detection decoder. Notably, Liu et al. [26] propose an alternative guidance strategy that aligns shallow detector features with downsampled enhanced images. Although these methodologies show performance improvements through auxiliary enhancement tasks, their guidance mechanisms remain fundamentally limited. Current approaches either utilize oversimplified feature alignment strategies or inadequately address the inherent divergence between restoration objectives and detection requirements. This critical oversight leaves substantial room for improvement in bridging the semantic gap between weather-invariant feature learning and task-aware detection optimization.

Addressing the aforementioned challenges, we propose **D**ual-**T**eacher **F**eature **A**lignment **(DTFA)** which redefines the paradigm of "clear feature alignment" for adverse weather detection. Our DTFA framework comprises four synergistic parts: the **I**nvariant **R**econstruction **T**eacher (IRT), the **S**emantic **P**rior **T**eacher (SPT), the adverse weather detection student, and two Adaptive Feature Bridging modules. Unlike conventional pixel-level restoration for alignment, our framework leverages complementary features from dual-teacher networks to guide the optimization of the lightweight adverse weather detection student through task-aware alignment. The IRT is derived from a pre-trained image reconstruction network, providing weather-invariant structural priors by aligning degraded inputs with clear counterparts. Meanwhile, the SPT, which is derived from a high-performance detection network pre-trained on clear images, offers task-aware semantic information to bridge the gap between reconstruction and detection objectives.

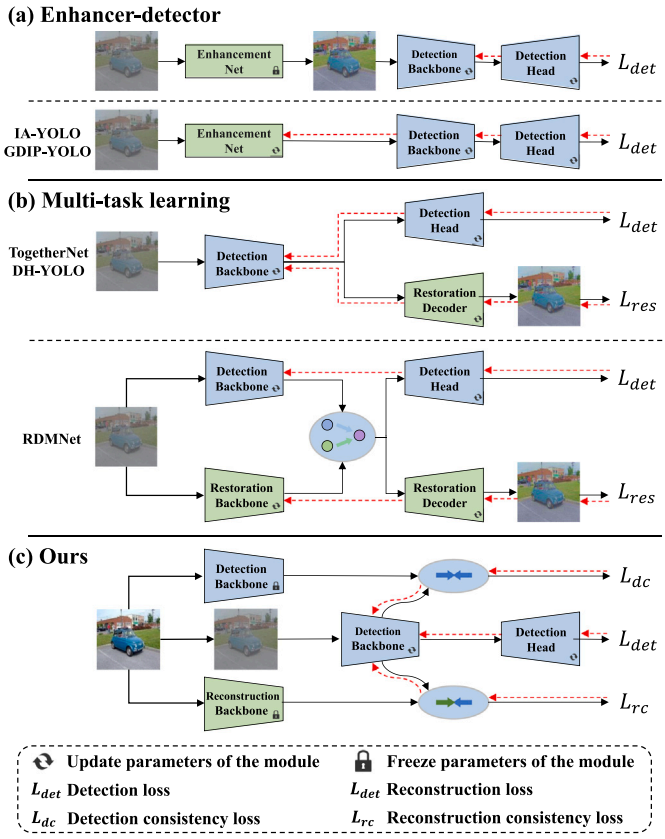


Fig. 1. The different paradigms of "clear image alignment". (a) Enhancer-detector pipelines that enhance (or restore) degraded images to align with their clear counterparts before detection networks. These pipelines employ a detection loss to update the parameters of the networks. (b) Multi-task learning frameworks that utilize a parallel restoration decoder to align with clear images. Both restoration loss and detection loss are employed to optimize the networks. (c) Our proposed DTFA framework, which aligns degraded images with their clear counterparts at the feature level through two pre-trained teacher networks. We utilize detection consistency loss, reconstruction consistency loss, and detection loss to train our student detection network.

Crucially, the adverse weather detection student, dynamically learns from integrated multi-scale features from both teachers via two Adaptive Feature Bridging modules. These modules enforce masked consistency constraints to resolve task discrepancies and calculate two consistency losses (reconstruction consistency loss and detection consistency loss). During inference, only the adverse weather detection student remains active, ensuring real-time performance without additional overhead.

Our main contributions can be summarized as follows:

- We propose a Dual-Teacher Feature Alignment (DTFA) framework that leverages an invariant reconstruction teacher (IRT) and a semantic prior teacher (SPT) to guide the adverse weather detection student. This design enables robust feature alignment without introducing additional computational overhead during inference.
- We develop Adaptive Feature Bridging modules that dynamically align multi-scale features between teachers and the student through masked consistency constraints. These modules resolve task discrepancies by fusing weather-invariant guidance and task-aware semantic information.
- Extensive experiments across single-condition scenarios, multi-weather environments, and real-world cross-domain validation demonstrate the robustness and practical viability of our framework in diverse adverse conditions.

2. Related works

2.1. Image restoration

In real-world scenarios, image quality is often adversely affected by bad weather. Image restoration serves as an effective solution to mitigate these degradations. As a crucial subfield of computer vision, image restoration focuses on recovering high-quality images from degraded observations. The field has evolved from prior-based, model-driven methods to deep learning-based, data-driven approaches. Early techniques often rely on hand-crafted priors (e.g., sparsity, self-similarity) and optimization algorithms; however, their ability to handle complex, real-world degradations remains limited.

The advent of convolutional neural networks (CNNs) marks a paradigm shift in image restoration by enabling end-to-end learning of mappings from degraded to clean images. Pioneering works such as SRCNN [27] demonstrate that even simple CNN architectures can achieve compelling results in super-resolution tasks. Subsequent research rapidly expanded to address more diverse and complex forms of degradation. For instance, specialized network architectures have been developed for specific weather-related phenomena: deraining methods [9,28,29] focus on removing rain streaks and accumulation; dehazing models such as EMPF-Net [14] and CINet [15] aim to eliminate haze effects through novel structural designs beyond conventional encoder-decoder frameworks; and desnowing techniques [16,17] tackle snowflake occlusion and accumulation—challenges often intensified by rapid motion or unstable imaging conditions.

Recognizing the impracticality of training separate models for each weather condition, recent studies have shifted their focus toward all-in-one frameworks for adverse weather removal. These methods aim to address multiple degradations—such as rain, haze, and snow—within a unified network architecture [9,10,30–34]. For example, AirNet [9] integrates a contrastive-based degraded encoder with a degradation-guided restoration network, enabling robust image recovery across various unknown corruption types and severity levels. RestorNet [10] introduces an improved contrastive learning method to design an efficient encoder-decoder network. Similarly, ADSM [32] introduces a self-prompting mechanism to guide the restoration process within a unified model. In comparison, DEMNet [33] advances this field with a multi-stage architecture, a novel contrastive loss for degradation extraction, and a spatial-channel integration block. Concurrently, DSRIR [34] introduces a progressive network with three subnets, dynamic patch attention, and uncertainty learning to handle diverse degradations, achieving superior performance with significantly fewer parameters.

Existing image restoration methods are capable of addressing many real-world degradation scenarios. When used as a pre-processing step for images captured under adverse weather conditions, these methods can mitigate visual impairments. However, preliminary research indicates that using image restoration independently as a pre-processing step may not yield optimal performance.

2.2. Object detection

Computer vision research has long prioritized object detection as a fundamental capability, driving sustained exploration in both academic and industrial settings [35]. The transformative impact of advancements in CNNs has established learning-based solutions as the predominant approach. Existing detection methodologies primarily adopt either region proposal-based or regression-based frameworks [36]. Models within the region proposal-based category, particularly those derived from R-CNN [37], emphasize accuracy improvements but frequently encounter speed limitations due to their sequential operational stages.

In contrast, regression-driven detectors like the YOLO family have transformed real-time object detection through unified single-stage localization and classification. YOLO's grid-based prediction mechanism enables the direct estimation of bounding boxes and class probabilities in a single pass, achieving remarkable speed-accuracy trade-offs. The YOLO series has evolved through key iterations: YOLOv4/v5 [38] introduced lightweight architectures for edge deployment, while YOLOv6 [39] and YOLOX [40] enhanced speed through reparameterization and anchor-free designs. Recent developments have focused on specialized variants such as YOLO-NAS (neural architecture search-optimized) [41], YOLO-R (robust to occlusions) [42], and ultra-lightweight YOLO-Tiny [43], which improve adaptability for real-time multimedia tasks, including drone-based surveillance and AR object interaction.

YOLO's efficiency makes it ideal for multimedia applications, including live video analytics, drone-based surveillance, and augmented reality, where low latency is critical. Although other regression-based methods, such as SSD [44] and RetinaNet [45], improve speed, they often fall short of YOLO in achieving a balance between accuracy and computational cost. Consequently, YOLO's architectural innovations enable its dominance in real-time detection, effectively addressing the limitations of region proposal-based approaches while maintaining competitive performance.

2.3. Object detection in adverse weather

Object detection in adverse weather conditions continues to be a persistent challenge in practical applications. Extensive research has been conducted to mitigate degradation caused by weather conditions. A straightforward approach involves employing image restoration techniques as pre-processing steps. For instance, Cui et al. [46] designed an image enhancer to pre-process low-light scene images. Additionally, Liu et al. [19] and Kalwar et al. [20] integrated a pre-processing module into image detection networks, effectively improving detection performance.

Recent studies have focused on multi-task learning and feature fusion strategies by incorporating image restoration as an auxiliary branch. TogetherNet [23] and DTARNet [47] optimize shared backbone networks through restoration module. DR-YOLO [22] integrates the atmospheric scattering model and object co-occurrence relations as prior knowledge to enable effective and efficient object detection in foggy conditions. Zhang et al. [24] introduced a feature filtering module in the restoration branch to eliminate redundant features. The RDMNet [25] dual-branch networks utilized multi-scale bidirectional feature fusion blocks to enhance the interaction between image detection and restoration features. Liu et al. [26] utilized downsampled enhanced images for feature alignment and guidance.

Furthermore, domain generalization for object detection has demonstrated potential in addressing variations in weather conditions. Pioneering work by Wu et al. [48] and Liu et al. [49] on single-source domain generalization has demonstrated superior model adaptability across diverse weather conditions. Notably, Jooyeon et al. [50] explored Vision-Language Models (VLMs) for open-vocabulary object detection, significantly enhancing generalization capabilities through their proposed methodology.

Building on these foundations, our work advances feature-level alignment through dual-teacher guidance, leveraging complementary clear-image features beyond pixel-level constraints to bridge reconstruction and detection objectives.

2.4. Knowledge distillation paradigms

Numerous studies have concentrated on knowledge distillation, with approaches pertinent to our work primarily categorized into three types: distillation for object detection, distillation for image restoration, and distillation with dual-teacher models.

2.4.1. Distillation for object detection

To address the challenges of low-quality face recognition, Zhang et al. [51] proposed the SA-TG method, which aligns the textures and attention of low-quality images with those of high-quality images using class-specific gradient textures. For enhancing object detectors through knowledge distillation, Yang et al. [52] introduced Focal and Global Distillation, which guide students to concentrate on critical regions and

global relationships learned from teachers. Zhang et al. [53] addressed efficient object detection in remote sensing images by employing an Adaptive Composite Feature Generation strategy. This approach improves the robustness of feature points under composite masks through adaptive feature mapping. To tackle feature misalignment, Liu et al. [54] developed a learnable nonlinear channel-wise transformation to align student–teacher features. To resolve the issues of foreground–background pixel imbalance and the lack of inter-pixel relation distillation, Zhang et al. [55] introduced two knowledge distillation frameworks: attention-guided distillation and non-local distillation.

2.4.2. Distillation for image enhancement

Hong et al. [56] proposed a knowledge-distilled dehazing network that leverages a pre-trained autoencoder as a teacher for image reconstruction, guiding the student dehazing network through process-oriented imitation learning. Lee et al. [57] introduced a distillation framework utilizing a teacher–student architecture, where the teacher network encodes degradation-aware features using ground-truth high-resolution images, and the student network, aligned with the FSRCNN's lightweight structure, reconstructs high-resolution images via feature distillation from the teacher's decoder.

2.4.3. Distillation with dual-teacher models

Zhou et al. [58] proposed a DTKD-Net that integrates a dynamic teaching strategy and contrastive learning. Their framework uses two teacher networks to guide the student network in preserving details in clear regions and restoring blurred areas, while a gating mechanism prevents over-reliance on the secondary teacher. Chen et al. [59] designed the Adaptive Modal Fusion Distillation framework, which employs a Modal Extraction Alignment module with focal-global attention mechanisms to align student and teacher modal features, enabling the student network to learn fusion strategies independently. Liu et al. [60] introduced Dual-Modeling Decouple Distillation, which decouples student features into normality and abnormality components and employs dual-modeling distillation to widen the gap between anomalous features and teacher representations, particularly at anomaly centers and edges.

Inspired by the remarkable distillation methods mentioned above, our work has expanded upon three key aspects. Firstly, our dual-teacher framework combines reconstruction and detection teachers, rather than relying on single-task models. Additionally, our framework emphasizes shallow feature alignment through multi-scale fusion, as opposed to traditional deep semantic alignment. Regarding the distillation strategy, our framework employs dynamic masking mechanisms and adaptive fusion to address task discrepancies, thereby avoiding direct feature imitation.

3. Method

3.1. Overall architecture

Based on the insight of leveraging clear images to guide the adverse weather detection and utilizing the student–teacher framework of knowledge distillation, we propose DTFA framework for object detection in challenging weather scenarios. The left part of Fig. 2 shows our DTFA framework, which comprises four components: the IRT, the SPT, the Adverse Weather Detection (AWD) student, and two Adaptive Feature Bridging (AFB) modules. The pre-training processes of the IRT and SPT are illustrated in the right part of Fig. 2. Notably, we retain only the shallow feature extraction layers to obtain our two teacher networks, which share the same network framework but have different weights. During the training process of the AWD student, the two teacher networks generate complementary clear features. Notably, the IRT and SPT share the same network framework as the AWD student, ensuring that the features are aligned. The AWD student learns these clear features through two AFB modules. The weights of the IRT and SPT are fixed, while the AWD student and two AFB modules are continuously optimized. During testing, only the AWD student remains. Therefore, no additional computational costs are incurred.

3.2. IRT-SPT teacher networks

To enable the AWD student to learn robust weather-invariant features and mitigate task gaps, we introduce the IRT and the SPT networks. The different training processes of two teacher networks make them pay more attention to the different aspects of the original images. Both the IRT and the SPT are pre-trained using clear images. During the training of the AWD student, clear images are input so that the features they provide can implicitly transfer clear information from the two teachers to the AWD student. In addition, the IRT is pre-trained using a reconstruction decoder and a reconstruction loss, which allows the IRT to focus more on global information that remains consistent across varying adverse weather conditions. On the contrary, the SPT is pre-trained with a detection head and a detection loss, which means its features are more conducive to the detection task. After pre-training, only the shallow layers (L1 and L2) are used as the IRT and SPT. During training, both of the IRT and SPT are fixed and generate complementary clear features.

3.2.1. Invariant reconstruction teacher (IRT)

Input images and target images of the reconstruction network are the same, facilitating the extraction of features at multiple scales while preserving the characteristics of the input images. Inspired by this concept, we propose a reconstruction network to pre-train our IRT. Considering that clear images are more salient and conducive to object detection in adverse weather conditions, we modify the input of the reconstruction network to include only clear images, while retaining these clear images as optimization targets. The MSE loss is utilized to optimize the parameters of the reconstruction network, which can be formalized as follows:

$$L_{rec} = \frac{1}{n}\sum_{i=1}^{n}(T_{REC}(Y_i) - Y_i)^2, \tag{1}$$

where n signifies the sample size, Y_i denotes clear images, with $T_{REC}(\cdot)$ denoting the reconstruction network. After training the reconstruction network, the shallow layers (L1 and L2) are used as our IRT to guide the training of AWD student.

During training of AWD student, the pre-trained IRT is fixed and provides robust and salient weather-invariant features T_{R1} and T_{R2}. Given the clear images Y_i, the process can be expressed as:

$$T_{R1} = CSP\left(W_{l1}^{3\times 3}(Focus(Y_i))\right), \tag{2}$$

$$T_{R2} = CSP^{\times 3}(W_{l2}^{3\times 3}(T_{R1})), \tag{3}$$

where $W_{l1}^{3\times 3}(\cdot)$ and $W_{l2}^{3\times 3}(\cdot)$ are 3×3 convolution layers, $CSP(\cdot)$ and $CSP^{\times 3}(\cdot)$ refer to Cross Stage Partial Networks [61] with different numbers of blocks, and $Focus(\cdot)$ refers to the focus layer [62].

The features T_{R1} and T_{R2} are then sent to the AFB module for the features from the AWD student, transferring the information from IRT to the AWD student.

3.2.2. Semantic prior teacher (SPT)

A common practice in teacher–student networks is knowledge distillation from high-performance teacher networks to low-performance student networks. Building upon this, we introduce SPT which is derived from a detection network pre-trained on clear images as the other teacher network. This detection network shares a same framework with the AWD student, which ensures the aligned features and efficient knowledge distillation. Unlike the previous clear-to-degraded knowledge distillation methods [51] that perform transfer at the final model outputs, our approach conducts clear-to-degraded feature transfer at the shallow-layer feature level.

Note that we primarily focus on the effectiveness of the DTFA architecture, which can be generalized to various detection networks.

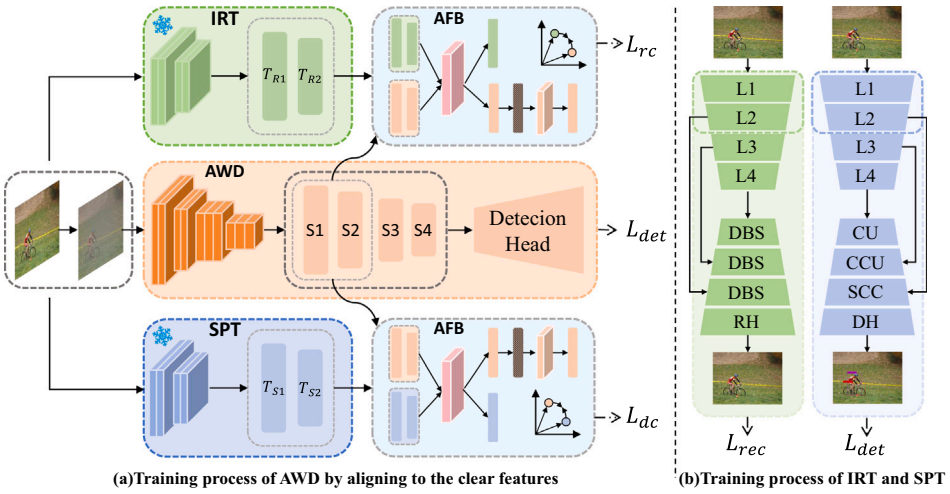


Fig. 2. Overview of our proposed architecture. Left: our proposed dual-teacher feature alignment is demonstrated. First, the Invariant Reconstruction Teacher (IRT) and Semantic Prior Teacher (SPT) generate complementary weather-invariant priors and task-aware semantic information features. Then, the Adverse Weather Detection (AWD) student learns by aligning to the clear features through two Adaptive Feature Bridging (AFB) modules. Only AWD student remains during inference. Right: The training process of IRT and SPT is shown, where clear images are utilized as the training data. The reconstruction network and detection network share the same structure of feature extraction layers, but employ different decoders and training loss. The L1 and L2 layers are utilized as the IRT and SPT in the trainning of AWD student.

Therefore, in this paper, we adopt a detection network design similar to that of RDMNet [25] for our pre-training of SPT and AWD student. In the training of the detection network, clear images are used as training data and a detection loss is employed, which can be expressed as:

$$L_{det} = L_{IoU} + L_{Cls} + L_{Focal}, \tag{4}$$

where L_{IoU}, L_{Cls}, and L_{Focal} refer to regression loss, classification loss, and focal loss [63], respectively. Similar to IRT, we select the shallow layers (L1 and L2) as our SPT to guide the training of AWD student.

During the training of AWD student, the pre-trained SPT is also fixed and provides task-aware semantic features T_{S1} and T_{S2} through a similar process described in (2) and (3). The features T_{S1} and T_{S2} from SPT complement the features T_{R1} and T_{R2} from IRT and are sent to another AFB module for the features from AWD student to align, transferring the information from SPT to AWD student.

3.3. Adverse weather detection (AWD) student

The AWD student is a detection network designed for images captured in adverse weather conditions. It serves as the primary optimization objective of our framework and is the sole component remaining for inference.

To preserve feature alignment and enable effective knowledge transfer, we generate training data for the AWD student by artificially degrading original clear images with synthetic weather-specific degradation patterns (*e.g.,* rain, fog and snow). This approach ensures strong paired consistency between the input images of teacher models and the AWD student while simultaneously addressing two critical challenges: 1) it circumvents the issue of collecting real-world degraded images by leveraging controllable synthetic degradation, and 2) it enables seamless extension to other weather conditions through diverse degradation patterns without necessitating additional data collection efforts.

Given a clear image Y_i, the synthetic degraded images Y_a can be generated by:

$$Y_a = Degradation(Y_i), \tag{5}$$

where $Degradation(\cdot)$ represents the specific synthetic weather-specific degradation pattern.

Inputting the Y_a to the AWD student, the AWD student first extracts two shallow features $S1$ and $S2$ using a similar manner as (2) and (3), which can be represented as follows:

$$S1 = CSP(W_{l1}^{3\times 3}(Focus(Y_a))), \tag{6}$$

$$S2 = CSP^{\times 3}(W_{l2}^{3\times 3}(S1)), \tag{7}$$

Next, the $S1$ and $S2$ are directed to two AFB modules to enforce consistency constraints, thereby optimizing the shallow feature extraction capabilities of the AWD student.

Besides, the AWD student produces the object detection result R_{det}, which is obtained by:

$$R_{det} = T_{AWD}(Y^a), \tag{8}$$

where $T_{AWD}(\cdot)$ refers to the AWD student. Consequently, the detection result R_{det} is compared to the labels to calculate the detection loss L_{det}, which is the same as (4).

3.4. Adaptive feature bridging (AFB) module

The AFB module is designed to effectively align the features from IRT, SPT, and AWD student, allowing the AWD student to learn efficiently from both IRT and SPT. However, due to the discrepancy in tasks, simply aligning features from one layer with the conventional design of the consistency loss function [26] cannot sufficiently guide the AWD student. Hence, our AFB module integrates multi-scale features and utilizes masked consistency constraints to further improve alignment. Considering that the structure of the two AFB modules connecting the two teachers and AWD student is the same, we introduce the AFB module that connects the IRT and the AWD student as an example in this section.

The architecture of our proposed AFB module is shown in Fig. 3. In order to capture multi-scale information, the features from IRT and AWD student are first fused. To preserve the distinctiveness between the IRT and the AWD student while effectively integrating their features, the features from both of them are processed through the fusion network using shared weights. For the features T_{R1} and T_{R2} from the IRT, as well as the features $S1$ and $S2$ from the AWD student, the feature fusion process can be expressed as follows:

$$F_R = W_{l1}^{1\times 1}(T_{R1}) + Upsample(W_{l2}^{1\times 1}(T_{R2})), \tag{9}$$

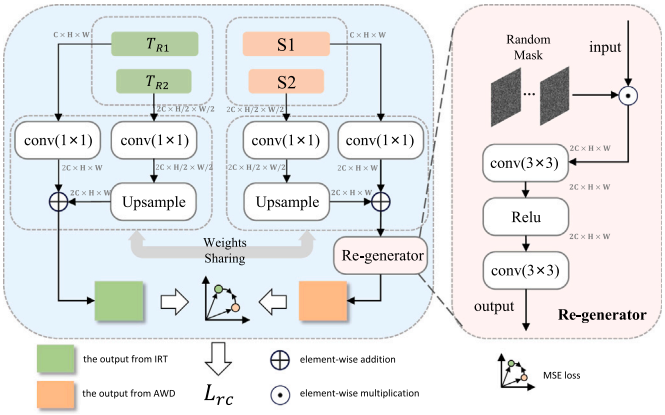


Fig. 3. The left side illustrates the overall architecture of the AFB module, which connects the IRT and the AWD student. The right side provides details of the re-generator, which employs a mask and convolutional layers to enhance the robustness of alignment.

$$F_S = W_{l1}^{1\times1}(S1) + Upsample(W_{l2}^{1\times1}(S2)), \tag{10}$$

where $W_{l1}^{1\times1}(\cdot)$ and $W_{l2}^{1\times1}(\cdot)$ refer to the 1×1 convolution layers with different channels, $Upsample(\cdot)$ refers to the upsampling operation, and F_R and F_S refer to the fusion results of the features from the IRT and the AWD student, respectively.

Inspired by masked autoencoders [64] and the insight of ensuring robust feature alignment, we propose a re-generator module. As shown in the right part of Fig. 3, a random mask is added to the input fused feature F_S and the convolutional layers compel it to generate features that closely resemble F_R, rather than simply mimicking F_R.

Firstly we generate a random mask, which can be formulated as:

$$M_{i,j} = \begin{cases} 0, & if \quad R_{i,j} < \lambda, \\ 1, & Otherwise. \end{cases} \tag{11}$$

where $R_{i,j}$ denotes a random value within $(0, 1)$, with indices i, j corresponding to the horizontal and vertical coordinates of the feature map, respectively. The hyperparameter λ, which controls the masked ratio, is empirically set to 0.65. Next, we apply the random mask to the feature F_S and try to generate a feature F_S^{gen}, which can be formulated as follows:

$$F_S^{gen} = W_{l2}^{3\times3}(Relu(W_{l1}^{3\times3}(M \cdot F_S))), \tag{12}$$

where $W_{l1}^{3\times3}(\cdot)$ and $W_{l2}^{3\times3}(\cdot)$ represent two 3×3 convolution layers, $Relu(\cdot)$ refers to activation layer, and M refers to the random mask. Finally, a MSE loss is employed to compute the reconstruction consistency loss L_{rc} between the AWD student and the IRT, which can be expressed as follows:

$$L_{rc} = \sum_{k=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(F_{S,k,i,j}^{gen} - F_{R,k,i,j}\right)^2, \tag{13}$$

where C, H, W denote the shape of the features F_S^{gen} and F_R. The L_{rc} encourages consistency between the AWD features and the IRT features, allowing the AWD student to learn the robust and salient weather-invariant characteristics of the IRT features.

Similar to the definition of L_{rc} described in (13), we obtain L_{dc} through the other AFB module that connects the AWD student and SPT. In this context, L_{dc} facilitates the alignment between the AWD features and the SPT features, thereby enabling AWD to acquire the task-specific characteristics inherent in the SPT features.

3.5. Total loss

The total loss of our model consists of three components: detection loss L_{det}, reconstruction consistency loss L_{rc}, and detection consistency loss L_{dc}. This can be defined as follows:

$$L_{total} = L_{det} + \lambda_{rc}L_{rc} + \lambda_{dc}L_{dc} \tag{14}$$

where λ_{rc} and λ_{dc} are loss weights. The optimal values of these two loss weights are determined through rigorous experimental validation, as detailed in Section 4.3.4. Here we set both λ_{rc} and λ_{dc} to 1×10^{-7}.

4. Experiments

4.1. Implementation details

We conduct experiments on our proposed DTFA framework utilizing the Pytorch framework. A reconstruction network and a detection network are first pre-trained using the VOC-Clean [68] dataset. We empirically train the two pre-trained networks for 100 epochs and select their shallow feature extraction components as our IRT and SPT. In the training process of AWD student, we freeze the IRT and SPT to ensure stability and continuity, while only the weights of AWD and AFB are updated. We train the AWD student detector using SGD with a batch size of 8 for 100 empirically determined epochs. The optimizer hyperparameters include an initial learning rate of 1×10^{-2}, momentum of 0.937, and a weight decay of 5×10^{-4}. A Cosine annealing scheduler dynamically adjusts the learning rate. Input images, for both training and inference, are uniformly resized to 640×640 pixels. Furthermore, the mosaic data augmentation strategy is not utilized to facilitate the processing of input images for the IRT. Unlike mainstream object detection algorithms or distillation methods, our proposed DTFA framework utilizes paired clear and degraded images as the supervisory signal for both the teacher and student networks. This approach provides clear features to the AWD student, thereby enhancing its performance in adverse weather conditions. We demonstrate the object detection performance of the proposed framework in adverse weather conditions, using images from rainy, foggy, and snowy scenarios as examples.

To evaluate the performance of the DTFA framework, we first conduct three separate experiments under rainy, foggy, and snowy conditions, respectively. We further validate the framework in adverse environments featuring combined weather conditions, where each image is degraded by two types of weather. Additionally, a multi-image multi-weather dataset comprising clear, rainy, foggy, and snowy images is used to assess detection performance across diverse weather scenarios. For methods whose training code is not open-sourced, we directly use the official weights provided by the authors for testing and mark these methods in *italics*. The remaining methods are re-trained and tested using our experimental setup. Details of the training and testing datasets are provided below.

4.1.1. Single rainy, foggy and snowy image datasets

To thoroughly assess the effectiveness of our method under diverse adverse weather conditions, we employ a benchmark that includes rainy, foggy, and snowy scenarios. Our experimental framework leverages three synthetic datasets (VOC-Rain, VOC-FOG, VOC-Snow) generated from the PASCAL VOC benchmark [68]. The synthetic datasets are constructed through physics-based degradation processes: VOC-Rain (comprising VOC-Rain-train and VOC-Rain-test) employs RainDS [65] for multi-scale rain streak synthesis with direction-aware rendering to VOC-Clean (9578 training/2129 inference images, divided into VOC-Clean-train/VOC-Clean-test) which preserves original VOC annotations. Similar to the VOC-Rain dataset, the VOC-FOG (comprising VOC-FOG-train and VOC-FOG-test) applies the atmospheric scattering model. Additionally, VOC-Snow (comprising VOC-Snow-train and VOC-Snow-test) integrates CSD's snow masking [67] with particle accumulation simulation. All the datasets mentioned above contain five classes for both training and testing: person, bicycle, car, motorbike, and bus.

4.1.2. Single-image multi-weather datasets

Real-world adverse weather conditions often occur not in isolation but as combinations, such as rain and fog or snow and fog. To simulate these compound weather scenarios, we have created two synthetic single-image multi-weather datasets: VOC-Rainfog and VOC-Snowfog. Each dataset contains images featuring a mixture of two weather types in every image, thereby enabling evaluation under combined degradations.

Specifically, VOC-Rainfog (comprising VOC-Rainfog-train and VOC-Rainfog-test) is synthesized by first applying the rain synthesis method from RainDS [65], followed by the introduction of fog using the atmospheric scattering model. Similarly, VOC-Snowfog (including VOC-Snowfog-train and VOC-Snowfog-test) is generated by first adding snow through the synthesis approach from CSD [67], and then applying the same fog simulation process. This two-step generation pipeline ensures that each image accurately reflects the combined visual effects of two weather conditions.

4.1.3. Multi-image multi-weather dataset

To facilitate unified model training across diverse weather conditions, we construct a multi-image, multi-weather dataset by combining the training sets of VOC-Clean [68], VOC-Rain, VOC-FOG, and VOC-Snow. The resulting mixed training set comprises 38,312 images representing clear, rainy, foggy, and snowy conditions. During testing, the trained model is evaluated separately on the four corresponding test sets—VOC-Clean-test, VOC-Rain-test, VOC-FOG-test, and VOC-Snow-test—to assess its mAP across each weather domain.

4.1.4. Real-world image dataset

To comprehensively assess the cross-domain generalization capability and real-world applicability of our proposed DTFA framework, we conduct evaluations across multiple real-world adverse weather conditions, including fog, rain, and snow.

We first utilize the widely adopted real-world foggy test dataset, RTTS [69]. The RTTS dataset comprises 4322 foggy images annotated with five object categories (i.e., person, bicycle, car, bus, and motorcycle), which correspond to those in the VOC-FOG dataset. This alignment enables a direct evaluation of object detection performance under real-world foggy conditions.

Given the scarcity of open-source, real-world rainy object detection datasets, we constructed a dedicated dataset, RealRain, to facilitate our evaluation. This dataset contains 120 real-world images captured under rainy conditions. We annotated two common and critical classes: person and car. RealRain enables a preliminary yet valuable assessment of our model's performance in rainy weather.

In contrast, building a large-scale annotated dataset for snowy conditions presents even greater challenges. Heavy snowfall is relatively infrequent, and data collection under such conditions is often hindered by safety concerns, limited accessibility, and severe visual degradation. Consequently, scenes heavily affected by snow are particularly difficult to capture, resulting in a notable scarcity of relevant samples. To partially address this issue, we compiled snow images from publicly available online sources. Although the limited number of snowy samples does not support robust quantitative evaluation, we provide qualitative detection results from our model. These visualizations offer practical insights into the model's behavior in snowy conditions, where snow accumulation can cause severe occlusion and appearance changes.

4.2. Experimental results

4.2.1. Results on rainy image dataset

We evaluate the performance of various methods under rainy conditions using the VOC-Rain dataset, with the results shown in Table 1. Specifically, the VOC-Rain-train dataset is used to train all models, while evaluation is conducted on the VOC-Rainy-test dataset. For comparison, we select the common object detection method YOLOXs [62] as the baseline. The models trained on clear images and rainy images are denoted as "YOLOXs* [62]" and "YOLOXs [62]", respectively. In addition, for the enhancer-detector pipelines, we adopt the image restoration methods AirNet [9] and RestorNet [10] as a pre-processing step before detection, with the models being AirNet-YOLOXs* [9] and RestorNet-YOLOXs* [10]. We also adopt MASFNet [21] as a novel enhancer-detector pipeline, where the parameters of both the restoration network and the detection network are updated using a detection loss. For the comparison of multi-task learning frameworks, we choose TogetherNet [23] and RDMNet [25].

As shown in Table 1, our method achieves a superior performance of 79.46% mAP, reflecting an increase of 0.69% mAP compared to the recent algorithm RDMNet [25]. In particular, our framework surpasses all previous methods across all categories of AP. Furthermore, Fig. 4 visually contrasts the detection outcomes on rainy images, thereby validating the proposed framework's ability to identify a greater number of objects with higher confidence scores. These experimental findings demonstrate the superiority and effectiveness of our DTFA framework for object detection under rainy conditions.

4.2.2. Results on foggy image dataset

To evaluate the performance of our DTFA framework in foggy weather conditions, we utilize the VOC-FOG dataset and compare our framework with several advanced object detection algorithms. For enhancer-detector pipelines, we select four mainstream image dehazing methods for pre-processing: DCP [11], AODNet [12], AECRNet [13], and RestorNet [10]. The previously mentioned image dehazing methods are integrated with YOLOXs* [62] to create DCP-YOLOXs*, AODNet-YOLOXs*, AECRNet-YOLOXs*, and RestorNet-YOLOXs*, respectively. Additionally, three end-to-end training enhancer-detector pipelines, DS-Net [66], IA-YOLO [19], and MASFNet [21], have been selected for a more comprehensive comparison. Similar to object detection in rainy weather conditions, TogetherNet [23] and RDMNet [25] are selected for the comparison of multi-task frameworks. The results of object detection in foggy scenarios for the aforementioned methods are illustrated in Table 2.

As indicated in Table 2, our DTFA framework achieves a superior performance of 80.45% mAP. Our framework achieves either the highest or the second-highest AP in all five categories. Notably, in the "Bus" category, the AP of our proposed framework increases by 1.74%. Additionally, Fig. 5 presents the visual results of object detection on images from VOC-FOG-test dataset, further validating the effectiveness of our proposed framework. Our framework can detect more objects with higher confidence in foggy scenarios, demonstrating the superiority of our DTFA framework.

4.2.3. Results on snowy image dataset

The VOC-Snow dataset is used to evaluate the performance of various methods in snowy scenarios. Specifically, the desnowing methods TKL [16] and LMQFormer [17] are selected as pre-processing steps before detection, resulting in the models named TKL-YOLOXs* and LMQFormer-YOLOXs*. In addition, we adopt MASFNet [21] as an end-to-end enhancer-detector method for comparison. As for multi-task learning frameworks, we include TogetherNet [23] and RDMNet [25] as comparison methods. The quantitative detection performance in snowy scenes is summarized in Table 3, with visual examples presented in Fig. 6.

As illustrated in Table 3, our proposed DTFA framework achieves superior performance in snowy weather conditions, with a 78.70% mAP. Particularly for "Bus" category, our framework demonstrates an improvement of 2.69% in AP compared to the highest AP achieved by previous methods. Furthermore, the results of various methods are visualized in Fig. 6, which also validate the superiority of our framework. These results demonstrate the superiority and effectiveness of the proposed method for processing snowy images.

Table 1
Object detection results on the VOC-Rain-test [65] dataset. "*" indicates the model is trained on clean images, whereas the methods without "*" are trained on images subjected to adverse weather conditions. For the tables shown in this paper, the best results are marked in **Bold**, and the second best are highlighted in Underline.

Method	Type	Class (AP (%))					mAP (%)
		Person	Bicycle	Car	Motorbike	Bus	
YOLOXs* [62]	Detector	75.94	63.00	66.92	65.32	73.10	68.86
YOLOXs [62]	Detector	80.98	65.88	74.54	70.83	84.35	75.32
AirNet-YOLOXs* [9]	Enhancer-Detector	80.06	67.81	70.75	69.87	82.77	74.25
RestorNet-YOLOXs* [10]	Enhancer-Detector	80.44	68.29	71.68	70.16	82.54	74.62
MASFNet [21]	Enhancer-Detector	76.25	51.61	63.76	55.67	73.66	64.19
TogetherNet [23]	Multi-Task	83.53	69.83	77.88	74.26	86.41	78.38
RDMNet [25]	Multi-Task	83.77	72.77	77.86	73.03	86.44	78.77
DTFA(Ours)	Dual-Teacher	**84.21**	**73.11**	**78.33**	**74.63**	**87.02**	**79.46**



Fig. 4. Object detection results of images captured in rainy conditions on the VOC-Rain-test [65] dataset.

4.2.4. Results on single-image multi-weather datasets

Real-world weather conditions are often complex and concurrent, involving mixtures such as rain-fog or snow-fog. To evaluate the robustness of our method under such compound degradations, we conduct experiments on the VOC-Rainfog and VOC-Snowfog datasets. We compare the proposed DTFA framework with several representative methods, including YOLOXs* [62], YOLOXs [62], MASFNet [21], TogetherNet [23], and RDMNet [25].

As summarized in Tables 4 and 5, our method achieves the highest overall mAP of 78.40% on VOC-Rainfog-test and 78.42% on VOC-Snowfog-test, demonstrating superior generalization capability in mixed weather scenarios. Its effectiveness is particularly evident in mixed snow-fog scenes, where it achieves the best AP in four out of five categories (person, car, motorbike, and bus), and ranks second in the bicycle category, indicating its effectiveness in handling complex visual degradations caused by overlapping snow and fog.

Qualitative results in Figs. 7 and 8 further confirm the advantages of our approach. Under compound weather conditions where multiple degradation factors coexist, DTFA detects more objects with higher confidence compared to competing methods, illustrating its practical strength in real-world adverse weather applications.

4.2.5. Results on multi-image multi-weather scenes

To further evaluate the object detection performance of our DTFA framework under multiple weather conditions, we conduct unified object detection experiments on four weather scene images, including clear images, rainy images, foggy images, and snowy images. In other words, we aim to train a unified model to handle images of the four weather

Table 2
Object detection results on the VOC-FOG-test [23] dataset. The methods in *italics* indicate that the evaluation uses the pre-trained model from the original paper without re-training. Thus, these results are directly comparable to those reported in the original work.

Method	Type	Class (AP (%))					mAP (%)
		Person	Bicycle	Car	Motorbike	Bus	
YOLOXs* [62]	Detector	73.09	57.22	69.55	59.83	77.34	67.41
YOLOXs [62]	Detector	83.28	67.67	77.75	68.91	81.70	75.86
DCP-YOLOXs* [11]	Enhancer-Detector	81.84	70.38	78.63	73.48	84.68	77.80
AODNet-YOLOXs* [12]	Enhancer-Detector	67.40	49.19	60.51	55.59	62.07	58.95
AECRNet-YOLOXs* [13]	Enhancer-Detector	80.47	67.82	76.97	72.46	82.73	76.09
RestorNet-YOLOXs* [10]	Enhancer-Detector	78.71	67.15	72.56	71.68	82.36	74.49
DS-Net [66]	Enhancer-Detector	72.44	60.47	**81.27**	53.85	61.43	65.89
IA-YOLO [19]	Enhancer-Detector	70.98	61.98	70.98	57.93	61.98	64.77
MASFNet [21]	Enhancer-Detector	77.20	50.89	65.75	55.04	74.57	64.69
DR-YOLO [22]	*Multi-Task*	*75.35*	*61.71*	*74.13*	*68.43*	***89.72***	*73.87*
TogetherNet [23]	Multi-Task	84.11	69.26	79.59	72.12	85.62	78.14
RDMNet [25]	Multi-Task	84.13	**73.41**	80.73	77.11	86.08	80.29
DTFA(Ours)	Dual-Teacher	**84.69**	71.14	81.25	**77.35**	87.82	**80.45**

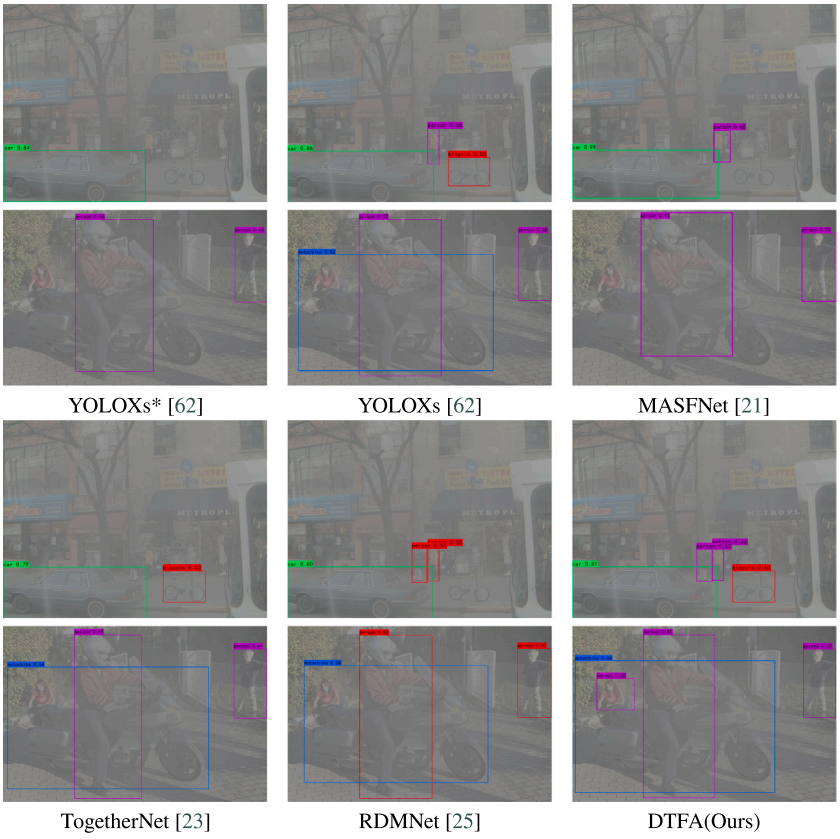


Fig. 5. Object detection results on images captured in foggy condition. They are from VOC-FOG-test [65] dataset.

Table 3
Object detection results on the VOC-Snow-test [67] dataset.

Method	Type	Class (AP (%))					mAP (%)
		Person	Bicycle	Car	Motorbike	Bus	
YOLOXs* [62]	Detector	78.40	64.88	70.80	56.90	81.15	70.43
YOLOXs [62]	Detector	81.16	66.64	75.43	70.87	83.28	75.48
TKL-YOLOXs* [16]	Enhancer-Detector	81.02	70.14	75.53	70.33	84.42	76.29
LMQFormer-YOLOXs* [17]	Enhancer-Detector	81.36	70.44	77.57	71.61	84.35	77.07
MASFNet [21]	Enhancer-Detector	76.52	52.04	64.38	55.98	75.33	64.85
TogetherNet [23]	Multi-Task	82.20	69.70	77.72	71.39	85.30	77.26
RDMNet [25]	Multi-Task	83.56	**70.71**	77.45	**74.12**	84.94	78.16
DTFA(Ours)	Dual-Teacher	**83.85**	69.09	**78.50**	74.09	**87.99**	**78.70**



Fig. 6. Object detection results of images captured in snowy condition on the VOC-Snow-test [65] dataset.

Table 4
Object detection results on the VOC-Rainfog-test dataset.

Method	Type	Class (AP (%))					mAP (%)
		Person	Bicycle	Car	Motorbike	Bus	
YOLOXs* [62]	Detector	67.74	49.18	62.46	27.20	71.74	50.66
YOLOXs [62]	Detector	81.30	63.48	74.68	67.77	86.13	74.67
MASFNet [21]	Enhancer-Detector	74.49	50.04	61.00	54.66	74.28	62.83
TogetherNet [23]	Multi-Task	82.02	70.53	75.10	67.81	**87.06**	76.50
RDMNet [25]	Multi-Task	82.35	**71.47**	76.84	73.62	85.45	77.95
DTFA(Ours)	Dual-Teacher	**84.70**	70.66	**78.38**	**74.13**	86.10	**78.40**

Table 5
Object detection results on the VOC-Snowfog-test dataset.

Method	Type	Class (AP (%))					mAP (%)
		Person	Bicycle	Car	Motorbike	Bus	
YOLOXs* [62]	Detector	73.00	58.25	69.04	44.30	78.07	64.53
YOLOXs [62]	Detector	79.81	58.74	73.63	66.28	81.69	72.03
MASFNet [21]	Enhancer-Detector	75.19	49.90	63.13	53.02	73.21	62.89
TogetherNet [23]	Multi-Task	82.51	66.49	76.48	74.49	84.46	76.89
RDMNet [25]	Multi-Task	82.97	**73.09**	76.98	73.29	84.34	78.13
DTFA(Ours)	Dual-Teacher	**83.07**	70.01	**77.06**	**76.01**	**85.94**	**78.42**

scenarios mentioned above. Specifically, we use our constructed mixed training dataset to train the model and test its performance in the above four weather scenes. We only train for 25 epochs in mixed datasets to fairly compare with the single weather scene. Only the multi-task methods are selected for comparison since the enhancer-detector pipelines are not suitable for handling multiple weather scene images.

The quantitative results on these four weather scenes are shown in Table 6. Notably, our method demonstrates superior performance across all the four weather scenes and achieves the highest Average mAP (79.25%), with an increase of 0.73% compared to the recent algorithm RDMNet [25]. The aforementioned results collectively underscore the exceptional object detection capability of the proposed method across a variety of weather conditions.

4.2.6. Cross-domain generalization on real-world scenes

To rigorously evaluate the cross-domain generalization capability and real-world applicability of our method, we conduct a comprehensive evaluation under the synthetic-to-real paradigm. All models are

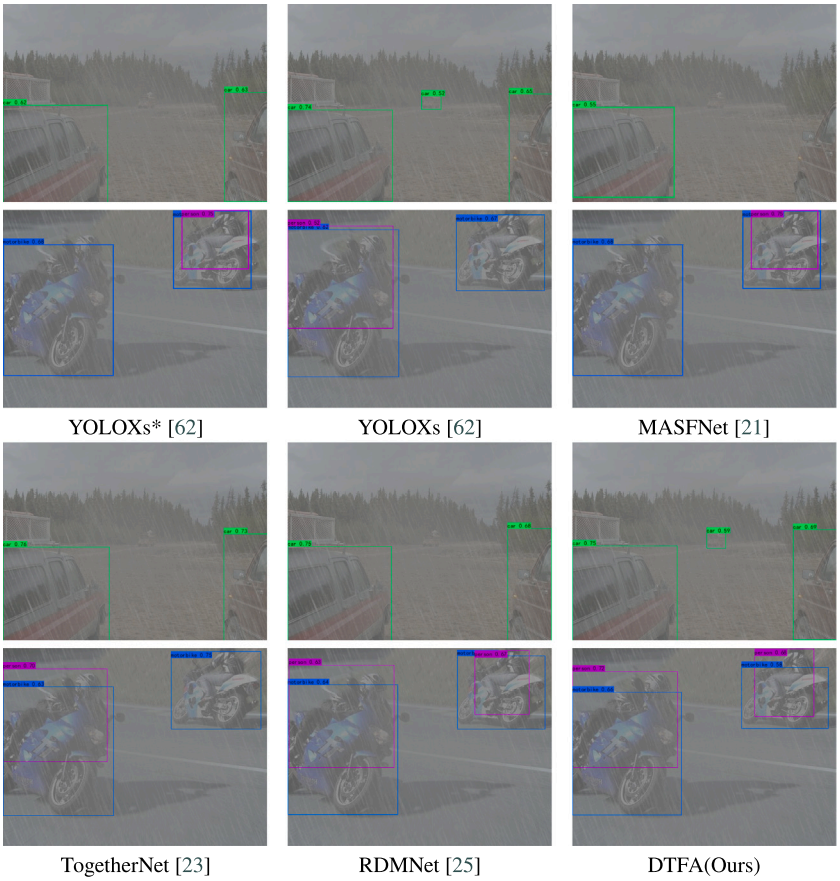


Fig. 7. Object detection performance on images under mixed rain-fog conditions from the VOC-Rainfog-test dataset.

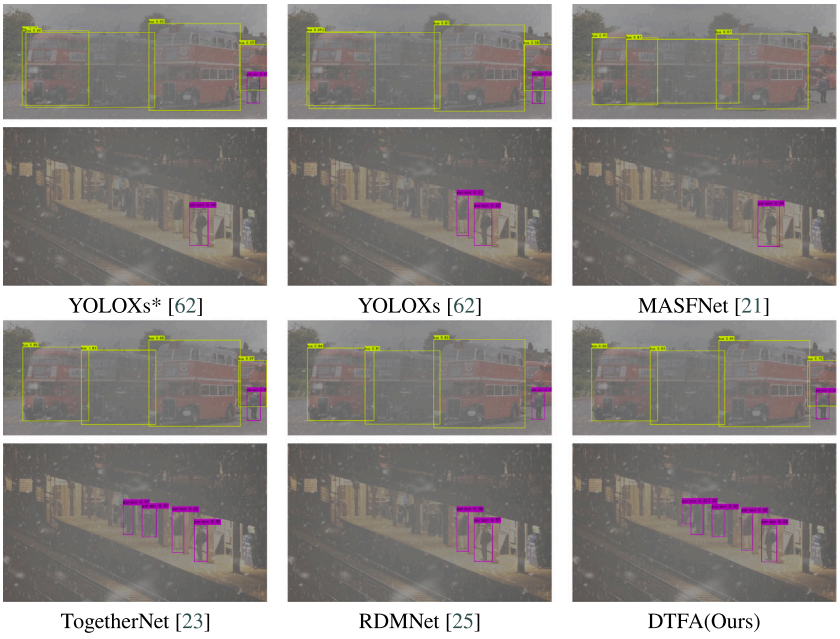


Fig. 8. Object detection performance on images under mixed snow-fog conditions from the VOC-Snowfog-test dataset.

trained exclusively on the synthetic dataset (VOC-FOG, VOC-Rain and VOC-Snow, respectively) and are subsequently tested directly on three types of real-world adverse weather images: foggy (RTTS [69]), rainy (RealRain), and snowy images.

As summarized in Table 7, our method achieves a mAP of 62.36%on the RTTS dataset, outperforming the strongest competitor RDMNet by 3.74%. Significant improvements are observed in the “Car”, “Motorbike”, and “Bus” categories, with gains of 7.09%, 7.12%, and

Table 6
Unified object detection results under multiple weather conditions. There are clean images in VOC-test, rainy images in VOC-Rain-test, foggy images in VOC-FOG-test, and snowy images in VOC-Snow-test.

Method	Type	Datasets/mAP(%)				Average
		VOC-test	VOC-FOG-test	VOC-Rain-test	VOC-Snow-test	
YOLOXs* [62]	Detector	79.09	68.86	67.41	70.43	71.45
YOLOXs [62]	Detector	75.77	74.50	75.50	74.24	75.00
MASFNet [21]	Enhancer-Detector	66.27	62.18	63.36	64.21	64.01
TogetherNet [23]	Multi-Task	78.86	76.74	76.78	77.03	77.35
RDMNet [25]	Multi-Task	79.48	78.19	78.72	77.69	78.52
DTFA(Ours)	Dual-Teacher	**80.30**	**79.39**	**78.81**	**78.51**	**79.25**

Table 7
Object detection results on the RTTS [69] of real-world foggy images. The methods in *italics* indicate that the evaluation uses the pre-trained model from the original paper without re-training. Thus, these results are directly comparable to those reported in the original work.

Method	Type	Class (AP (%))					mAP (%)
		Person	Bicycle	Car	Motorbike	Bus	
YOLOXs* [62]	Detector	80.28	50.75	68.23	41.89	28.89	54.01
YOLOXs [62]	Detector	81.78	56.70	70.23	49.48	31.57	57.95
DCP-YOLOXs* [11]	Enhancer-Detector	81.16	51.34	71.13	47.20	31.09	56.38
AODNet-YOLOXs* [12]	Enhancer-Detector	78.16	44.49	65.54	38.88	25.57	50.53
AECRNet-YOLOXs* [13]	Enhancer-Detector	80.85	52.75	68.60	46.29	30.43	55.78
RestorNet-YOLOXs* [10]	Enhancer-Detector	77.48	51.43	60.92	43.12	29.16	52.42
DS-Net [66]	Enhancer-Detector	68.81	18.02	46.13	15.15	15.44	32.71
IA-YOLO [19]	Enhancer-Detector	67.25	35.28	41.14	20.97	13.64	35.66
MASFNet [21]	Enhancer-Detector	70.78	41.43	57.34	28.59	23.31	44.29
DR-YOLO [22]	*Multi-Task*	*73.28*	***57.95***	*68.04*	*52.96*	*38.34*	*58.12*
TogetherNet [23]	Multi-Task	81.14	54.12	72.39	47.02	32.99	57.53
RDMNet [25]	Multi-Task	**82.44**	53.58	73.08	49.45	34.58	58.62
DTFA(Ours)	Dual-Teacher	82.29	57.87	**76.04**	**56.57**	**39.04**	**62.36**

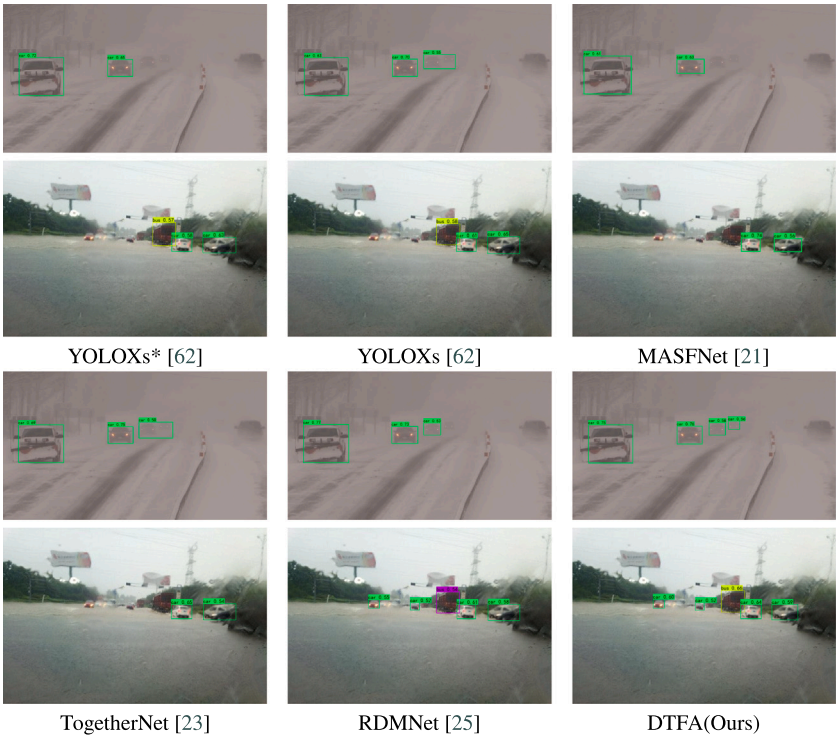


Fig. 9. Cross-domain generalization object detection results on real-world foggy image RTTS [69] dataset.

4.46%, respectively. Qualitative results in Fig. 9 further confirm that our method detects more objects with higher confidence in dense fog conditions.

On RealRain dataset, DTFA achieves the best mAP of 66.46% as shown in Table 8. It notably achieves the highest AP for the “Car” class and the second-best for “Person”, demonstrating strong robustness to

Table 8
Object detection results on the RealRain dataset of real-world rainy images.

Method	Type	Class (AP (%))		mAP (%)
		Person	Car	
YOLOXs* [62]	Detector	53.89	60.74	57.35
YOLOXs [62]	Detector	60.63	57.08	58.86
MASFNet [21]	Enhancer-Detector	**76.28**	46.34	61.61
TogetherNet [23]	Multi-Task	65.70	55.78	60.74
RDMNet [25]	Multi-Task	67.49	55.82	61.66
DTFA(Ours)	Dual-Teacher	69.97	**62.95**	**66.46**

rain streaks and visibility degradation. Visual comparisons in Fig. 10 illustrate that our method maintains more stable detection performance under rainfall.

Although quantitative annotations are not available for snowy scenes, the detection visualizations in Fig. 11 provide qualitative evidence of our method's effectiveness. Under snowfall and accumulated snow conditions, DTFA yields more complete and confident detections compared to other methods, showing better resistance to snow-induced occlusions and appearance variations.

The consistent superiority of our framework across fog, rain, and snow scenarios—under a strict synthetic-to-real evaluation protocol—strongly validates its cross-domain generalization capability and practical utility in diverse real-world adverse weather conditions.

4.2.7. Analysis of computational complexity

To assess the practical applicability of our DTFA framework, we evaluate the computational complexity during inference across different detection methods by measuring the number of model parameters and Floating Point Operations (FLOPs) at a standardized input resolution of 640×640.

As shown in Table 9, using the same detector, DTFA employs 14.11 million parameters, matching the efficiency of TogetherNet [23] while being significantly more compact than RDMNet, which uses 22.32 million parameters [21]. This demonstrates that our method maintains the base detector's parameter efficiency without the overhead of additional modules. Such efficiency directly enhances DTFA's suitability for real-world deployment.



Fig. 10. Cross-domain generalization object detection results on real-world rainy image RealRain dataset.

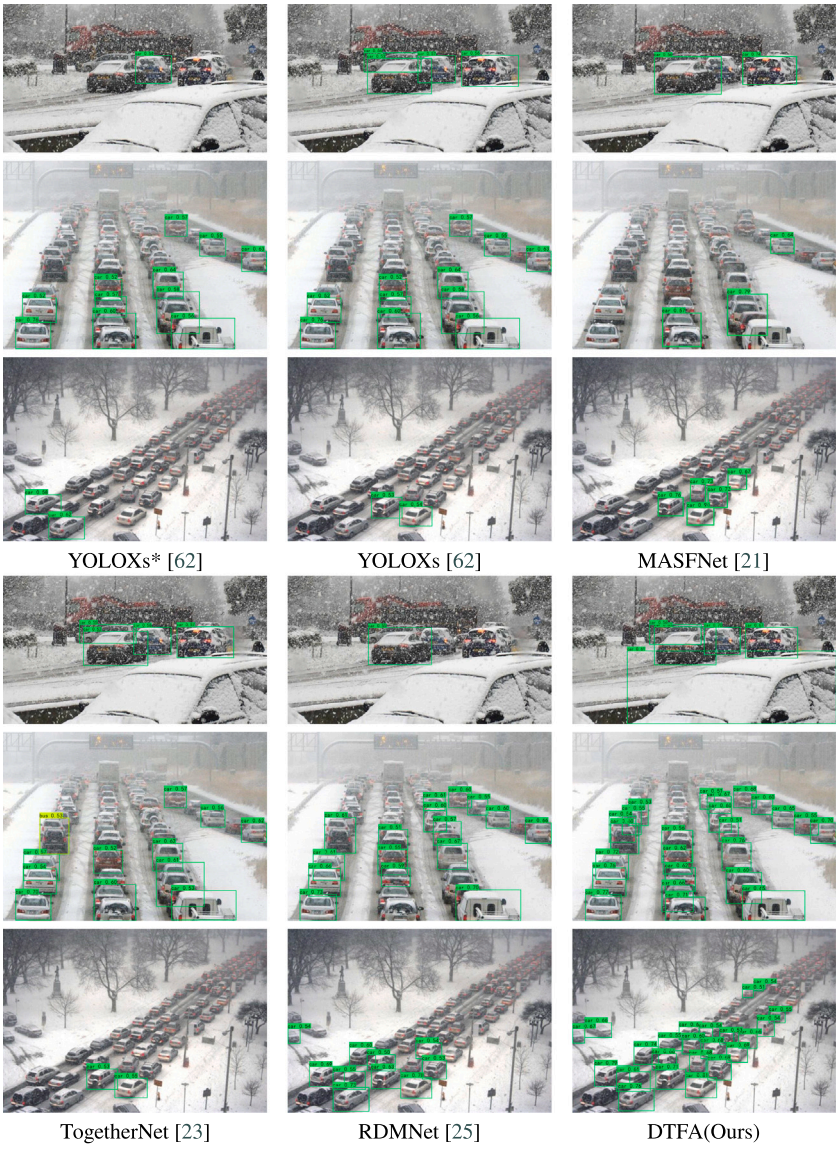


Fig. 11. Cross-domain generalization object detection results on real-world snowy images.

Enhancer-detector methods such as AECRNet-YOLOXs (309.31 GFLOPs) and AirNet-YOLOXs (1896.39 GFLOPs) exhibit high computational demands, primarily due to the overhead of their integrated restoration networks. In contrast, our method requires only 17.82 GFLOPs, representing a significant reduction compared to RDMNet (25.37 GFLOPs). This improvement is achieved by eliminating the need to fuse features from a separate restoration network during inference. This streamlined architecture highlights the efficiency of our design.

Consequently, DTFA is significantly more efficient than methods that incorporate restoration networks and maintains a leaner profile compared to other advanced detection frameworks, underscoring its advantages for practical deployment.

Table 9
Comparison of the computational complexity for various detection methods. The image size for calculating FLOPs is 640×640.

Model	Parameters(M)	FLOPs(G) ↓
YOLOXs [62]	8.96	13.38
AECRNet-YOLOXs [13]	11.57	309.31
AirNet-YOLOXs [9]	18.22	1896.39
TKL-YOLOXs [16]	37.67	166.77
MASFNet [21]	6.02	12.66
TogetherNet [23]	14.11	17.29
RDMNet [25]	22.32	25.37
DTFA(Ours)	14.11	17.82

4.3. Ablation study

4.3.1. Two-teacher architecture analysis

To validate the necessity of the two-teacher design, we conduct ablation studies on the framework. For the sake of convenience, all ablation experiments presented in this paper are conducted by training on VOC-FOG-train dataset and testing on VOC-FOG-test dataset. As summarized in Table 10, four experimental configurations are compared: N1 is the baseline AWD student trained solely with L_{det}; N2 is the AWD student guided by SPT; N3 is the AWD student guided by IRT; and N4 is our full framework that integrates both SPT and IRT. The N1 baseline achieves 76.82% mAP, while introducing either SPT in N2 or IRT in N3 improves performance to 79.75% and 80.00%, with respective increases of 2.93% and 3.18% in mAP. This demonstrates that both teachers independently

Table 10
Ablation study of different training frameworks on VOC-FOG-test dataset.

Model	N1	N2	N3	N4
SPT	w/o	✓	w/o	✓
IRT	w/o	w/o	✓	✓
mAP(%)	76.82	79.75	80.00	**80.45**

Table 11
Ablation study of aligned features.

Model	M1	M2	M3	M4	M5	M6	M7	M8	M9
F1	w/o	✓	w/o	w/o	w/o	✓	w/o	✓	✓
F2	w/o	w/o	✓	w/o	w/o	✓	w/o	✓	✓
F3	w/o	w/o	w/o	✓	w/o	w/o	✓	✓	✓
F4	w/o	w/o	w/o	w/o	✓	w/o	✓	w/o	✓
mAP(%)	76.82	79.71	79.02	78.9	79.21	**80.45**	79.23	79.32	79.46

enhance detection performance: SPT injects task-specific semantic priors, while IRT provides weather-invariant clarity through reconstruction guidance.

Crucially, the N4 model, which combines both teachers, achieves the highest mAP of 80.45%, surpassing the performance of individual teacher configurations. This demonstrates the complementary nature of the dual guidance mechanisms, where SPT preserves category-discriminative features, while IRT recovers fine-grained spatial details. Together, they synergistically address feature corruption caused by weather conditions. These results conclusively validate the architectural effectiveness of the proposed DTFA framework.

4.3.2. Ablation study on feature combinations

To enhance detection performance in degraded images, we investigate the effects of various aligned features. Table 11 presents the results of four aligned features (F1-F4), derived from the L1-L4 layers, respectively. Our experiments assess several distinct and significant feature combinations (M1-M9). M1 serves as the baseline model, trained without any feature guidance. For models guided by a single feature (M2–M5), we directly compute the consistency loss between the teacher networks using the re-generator module, without feature fusion. For models guided by multi-scale features (M6–M9), the features are first fused through convolutional layers and upsampling, after which the consistency loss is computed using the same re-generator module.

As shown in Table 11, the M2-M9 methods outperform the M1 method, highlighting the effectiveness of feature guidance. The M6 method achieves the highest mAP of 80.45%, demonstrating that distilling the F1 and F2 features from shallow feature extraction layers leads to better detection performance, a phenomenon consistent with previous studies [26]. In comparison to the M6 method, the M8 and M9 methods incorporate additional F3 or F4 features but yield inferior performance. This may be attributed to the M6 method's features containing more relevant information, making them more suitable for alignment, while the M8 and M9 methods introduce deeper, less effective semantic information. Consequently, we align the feature maps from the two shallow layers, L1 and L2, and calculate the consistency loss.

4.3.3. The impact of alignment modules

We systematically evaluate the synergistic effects of mask mechanism and convolutional layer configurations employed in our AFB module through ten controlled ablation studies. Our experiments strictly control variables, testing the effects of different convolutional layer counts

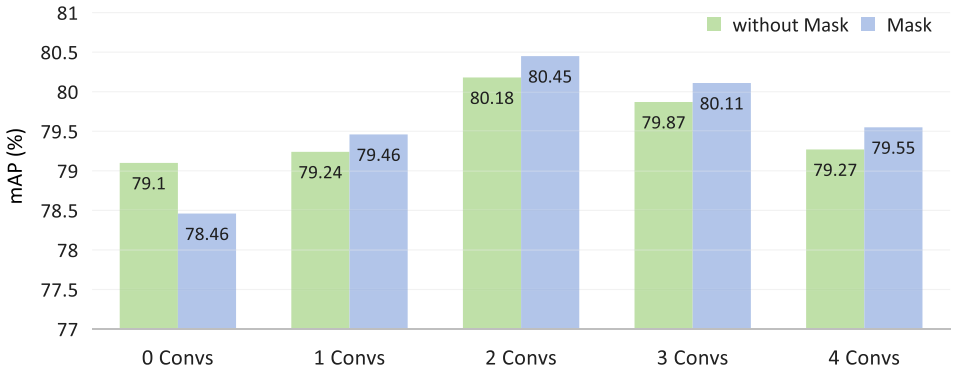


Fig. 12. The mAP results for the ablation study of various alignment methods.

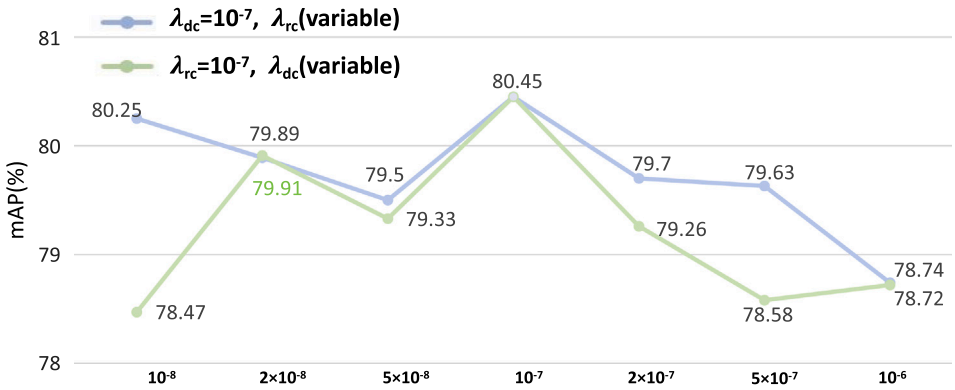


Fig. 13. Ablation study on the reconstruction consistency loss L_{rc} and detection consistency loss L_{dc} (loss weight λ_{rc} and λ_{dc}).

(0-4 layers) across two groups: one without a mask and one with a mask.

As illustrated in Fig. 12, we assess the effectiveness of the masks. With the same number of convolutional layers, models that incorporate a mask generally demonstrate superior performance compared to those that do not. For instance, using the same two convolutional layers, the model trained with masks achieves an mAP of 80.45%, significantly surpassing the model that lacks masks, which has an mAP of 80.18%. This trend underscores the effectiveness of the masks in enhancing feature alignment.

In addition, regardless of whether masks are applied, two layers of convolution exhibit the best performance. When the number of convolution layers exceeds two (*e.g.,* three to four layers), model performance declines, suggesting that an excessive number of layers may lead to overfitting and diminish the effectiveness of knowledge distillation, resulting in reduced global alignment capability. Conversely, when the number of convolution layers is zero or one, model performance also experiences a significant decline, indicating that too few layers may lack the capacity for feature reconstruction, which limits performance improvement. Moreover, the experimental results of the baseline control further demonstrate the effectiveness of our aligned method. The baseline model, which lacks both convolutional layers and a mask, exhibits the lowest performance at 79.10%. In contrast, incorporating either masks or convolutional layers enhances performance, and incorporating both masks and convolutional layers achieves even better results, thereby underscoring the necessity of their combined design.

As mentioned above, the combination of the masking mechanism and two-layer convolution achieves an optimal balance between feature alignment and generalization ability, thereby maximizing the effectiveness of knowledge distillation.

4.3.4. The impact of weights on two consistency loss functions

To improve detection performance in challenging weather conditions, our DTFA framework utilizes a loss function that contains L_{det}, L_{rc}, and L_{dc}. Accordingly, two loss weights (λ_{rc} and λ_{dc}) are utilized to balance the effects of the teacher networks. Extensive experiments are conducted using the VOC-FOG-test dataset to optimize the values of λ_{rc} and λ_{dc}, as illustrated in Fig. 13. As shown, when $\lambda_{rc} = 1e-7$ and $\lambda_{dc} = 1e-7$ are set, the performance of DTFA is maximized.

5. Conclusion

In this paper, we propose a dual-teacher feature alignment (DTFA) framework for object detection in adverse weather conditions. This framework employs two teachers (IRT and SPT) derived from a reconstruction network and a detection network that have been pre-trained on clear images, along with an adverse weather detection student. The IRT and SPT provide complementary weather-invariant and task-aware semantic information to guide the adverse weather detection student. Furthermore, two adaptive feature bridging modules are utilized to dynamically align multi-scale features between the two teachers and the student detector, effectively addressing task discrepancies through masked consistency constraints. The adverse weather detection student learns from the complementary clear features that are generated by IRT and SPT and enhances the detection performance in challenging weather scenarios. During testing, only the adverse weather detection student remains. Therefore, no additional computational costs are incurred. Extensive experiments conducted under rainy, foggy, and snowy conditions demonstrate the effectiveness and superiority of our DTFA framework compared to existing methods, as well as its generalization capabilities and applicability in real-world scenarios.

CRediT authorship contribution statement

Rui Hu: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hanjun Zheng:** Writing – review & editing, Visualization, Validation, Investigation, Formal analysis. **Shengjie Ye:** Writing – review & editing, Visualization, Validation, Investigation, Formal analysis. **Linbo Qing:** Resources, Funding acquisition. **Honggang Chen:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] P. Li, X. Yuan, H. Xu, J. Wang, Y. Wang, Focus DETR: focus detection transformer for ship wall-climbing robot real-time object detection, Neurocomputing 651 (2025) 131049. ISSN 0925-2312.
[2] K. Oksuz, B. Cam, S. Kalkan, E. Akbas, Imbalance problems in object detection: a review, IEEE Trans. Pattern Anal. Mach. Intell. 43 (8) (2020) 3382–3415.
[3] Y. Zhang, J. Wen, C. Li, Bi-directional domain adaptation for cross-domain HSI classification, IEEE Trans. Circuits Syst. Video Technol. 35 (3) (2025) 1120–1134.
[4] C. Gong, H. Dong, G. Yang, P. Yang, A weakly supervised salient object detection framework based on structured scribble, Neurocomputing 652 (2025) 131057. ISSN 0925-2312.
[5] W. Zhou, Q. Guo, J. Lei, L. Yu, J.-N. Hwang, Irfr-net: interactive recursive feature-reshaping network for detecting salient objects in rgb-d images, IEEE Trans. Neural Netw. Learn. Syst. 36 (3) (2025) 4132–4144.
[6] Y. Li, F. Liu, Y. Qin, L. Dai, L. Zhang, W. Xie, Small object detection network based on progressive enhanced multi-level feature fusion, Neurocomputing 649 (2025) 130755. ISSN 0925-2312.
[7] C. Yan, X. Chang, M. Luo, H. Liu, X. Zhang, Q. Zheng, Semantics-guided contrastive network for zero-shot object detection, IEEE Trans. Pattern Anal. Mach. Intell. 46 (3) (2024) 1530–1544.
[8] B. Song, J. Chen, W. Liu, J. Fang, Y. Xue, X. Liu, Yolo-elwnet: a lightweight object detection network, Neurocomputing 636 (2025) 129904. ISSN 0925-2312.
[9] B. Li, X. Liu, P. Hu, Z. Wu, J. Lv, X. Peng, All-in-one image restoration for unknown corruption, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2022, pp. 17452–17462.
[10] X. Wang, H. Chen, H. Gou, J. He, Z. Wang, X. He, L. Qing, R.E. Sheriff, Restornet: an efficient network for multiple degradation image restoration, Knowl.-Based Syst. 282 (2023) 111116.
[11] K. He, J. Sun, X. Tang, Single image haze removal using dark channel prior, IEEE Trans. Pattern Anal. Mach. Intell. 33 (12) (2010) 2341–2353.
[12] B. Li, X. Peng, Z. Wang, J. Xu, D. Feng, Aod-net: all-in-one dehazing network, in: Proc. IEEE Int. Conf. Comput. Vis, 2017, pp. 4770–4778.
[13] H. Wu, Y. Qu, S. Lin, J. Zhou, R. Qiao, Z. Zhang, Y. Xie, L. Ma, Contrastive learning for compact single image dehazing, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2021, pp. 10551–10560.
[14] Y. Wen, T. Gao, J. Zhang, Z. Li, T. Chen, Encoder-free multiaxis physics-aware fusion network for remote sensing image dehazing, IEEE Trans. Geosci. Remote Sens. 61 (2023) 1–15.
[15] Y. Wen, T. Gao, T. Chen, Z. Li, M. Liu, L. Liu, Cross-level interaction and intra-level fusion network for remote sensing image dehazing, IEEE Trans. Geosci. Remote Sens. 63 (2025) 1–14.
[16] W.-T. Chen, Z.-K. Huang, C.-C. Tsai, H.-H. Yang, J.-J. Ding, S.-Y. Kuo, Learning multiple adverse weather removal via two-stage knowledge learning and multi-contrastive regularization: toward a unified model, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2022, pp. 17653–17662.
[17] J. Lin, N. Jiang, Z. Zhang, W. Chen, T. Zhao, Lmqformer: a laplace-prior-guided mask query transformer for lightweight snow removal, IEEE Trans. Circuits Syst. Video Technol. 33 (11) (2023) 6225–6235.

[18] S. Xu, M. Zhang, W. Song, H. Mei, Q. He, A. Liotta, A systematic review and analysis of deep learning-based underwater object detection, Neurocomputing 527 (2023) 204–232.
[19] W. Liu, G. Ren, R. Yu, S. Guo, J. Zhu, L. Zhang, Image-adaptive YOLO for object detection in adverse weather conditions, in: Proc. AAAI Conf. Artif. Intell, vol. 36, 2022, pp. 1792–1800.
[20] S. Kalwar, D. Patel, A. Aanegola, K.R. Konda, S. Garg, K.M. Krishna, Gdip: gated differentiable image processing for object detection in adverse conditions, in: Proc. IEEE Int. Conf. Robot. Autom, 2023, pp. 7083–7089.
[21] Z. Liu, T. Fang, H. Lu, W. Zhang, R. Lan, Masfnet: multiscale adaptive sampling fusion network for object detection in adverse weather, IEEE Trans. Geosci. Remote Sens. 63 (2025) 1–15.
[22] F. Zhong, W. Shen, H. Yu, G. Wang, J. Hu, Dehazing & reasoning YOLO: prior knowledge-guided network for object detection in foggy weather, Pattern Recogn. 156 (2024) 110756.
[23] Y. Wang, X. Yan, K. Zhang, L. Gong, H. Xie, F.L. Wang, M. Wei, Togethernet: bridging image restoration and object detection together via dynamic enhancement learning, in: Comput. Graph. Forum, vol. 41, Wiley Online Library, 2022, pp. 465–476.
[24] K. Zhang, X. Yan, Y. Wang, J. Qi, Adaptive dehazing YOLO for object detection, in: Proc. Int. Conf. Artif. Neural Netw, Springer, 2023, pp. 14–27.
[25] X. Wang, X. Liu, H. Yang, Z. Wang, X. Wen, X. He, L. Qing, H. Chen, Degradation modeling for restoration-enhanced object detection in adverse weather scenes, IEEE Trans. Intell. Veh. 10 (2024) 2064–2079.
[26] H. Liu, F. Jin, H. Zeng, H. Pu, B. Fan, Image enhancement guided object detection in visually degraded scenes, IEEE Trans. Neural Netw. Learn. Syst. 35 (10) (2023) 14164–14177.
[27] C. Dong, C.C. Loy, K. He, X. Tang, Image super-resolution using deep convolutional networks, IEEE Trans. Pattern Anal. Mach. Intell. 38 (2) (2015) 295–307.
[28] X. Chen, J. Pan, J. Dong, Bidirectional multi-scale implicit neural representations for image deraining, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024, pp. 25627–25636.
[29] Y. Yang, Y. Zhang, Z. Cui, H. Zhao, T. Ouyang, Single image deraining using scale constraint iterative update network, Expert Syst. Appl. 236 (2024) 121339.
[30] Y. Wan, Y. Cheng, M. Shao, C. Wang, Rdm-ir: task-adaptive deep unfolding network for all-in-one image restoration, Knowl.-Based Syst. 304 (2024) 112543. ISSN 0950-7051, https://doi.org/10.1016/j.knosys.2024.112543, https://www.sciencedirect.com/science/article/pii/S0950705124005337
[31] T. Gao, Y. Wen, K. Zhang, J. Zhang, T. Chen, L. Liu, W. Luo, Frequency-oriented efficient transformer for all-in-one weather-degraded image restoration, IEEE Trans. Circuits Syst. Video Technol. 34 (2024) 1886–1899, https://doi.org/10.1109/TCSVT.2023.3299324.
[32] Y. Wen, T. Gao, Z. Li, J. Zhang, K. Zhang, T. Chen, All-in-one weather-degraded image restoration via adaptive degradation-aware self-prompting model, IEEE Trans. Multim. 27 (2025) 3343–3355.
[33] Y. Yang, X. Wang, X. Lin, H. Chen, Demnet: a degradation difference enabled multi-stage network for multiple degradation image restoration, Knowl.-Based Syst. 318 (2025) 113426, https://doi.org/10.1016/j.knosys.2025.113426
[34] X. Wang, X. Liu, Y. Yang, Z. Wang, X. He, H. Chen, Y. Li, P. Wang, Dsrir: dynamic spatial refinement learning for progressive all-in-one image restoration, Inf. Process. Manag. (2025) 104380. ISSN 0306-4573, https://doi.org/10.1016/j.ipm.2025.104380.
[35] L. Liu, W. Ouyang, X. Wang, P. Fieguth, J. Chen, X. Liu, M. Pietikäinen, Deep learning for generic object detection: a survey, Int. J. Comput. Vis. 128 (2) (2020) 261–318.
[36] J.R.R. Uijlings, K.E.A. van de Sande, T. Gevers, A.W. Smeulders, Selective search for object recognition, Int. J. Comput. Vis. 104 (2) (2013) 154–171.
[37] R. Girshick, J. Donahue, T. Darrell, J. Malik, Rich feature hierarchies for accurate object detection and semantic segmentation, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2014, pp. 580–587.
[38] A. Bochkovskiy, C.-Y. Wang, H.-Y. Liao, Yolov4: Optimal speed and accuracy of object detection, arXiv preprint arXiv:2004.10934, 2020.
[39] Z. Ge, S. Liu, F. Wang, Yolov6: a single-stage object detection framework for industrial applications, IEEE Conf. Comput. Vis. Pattern Recognit. (2021) 12429–12438.
[40] Z. Tian, Y. Jiang, X. Wang, Yolox: exceeding YOLO series in 2021, IEEE Int. Conf. Comput. Vis. (2021) 958–967.
[41] W. Chen, L. Wang, H. Zhang, M. Li, J. Zhou, Yolo-nas: neural architecture search-optimized object detection for real-time edge computing, IEEE Trans. Neural Netw. Learn. Syst. 35 (5) (2024) 2245–2258.
[42] Y. Liu, X. Zhang, W. Chen, Q. Li, Yolo-r: robust occlusion-aware object detection for autonomous driving, IEEE Trans. Image Process. 32 (2023) 7895–7908.
[43] L. Zhang, Y. Wang, X. Li, B. Zhou, Yolo-tiny: ultra-lightweight object detection for resource-constrained embedded systems, IEEE Trans. Multimedia 24 (2022) 4321–4333.
[44] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, A.C. Berg, SSD: single shot multibox detector, in: Proc. Eur. Conf. Comput. Vis, Springer, 2016, pp. 21–37.
[45] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proc. IEEE Int. Conf. Comput. Vis, 2017, pp. 2980–2988.
[46] X. Cui, L. Ma, T. Ma, J. Liu, X. Fan, R. Liu, Trash to treasure: low-light object detection via decomposition-and-aggregation, in: Proc. AAAI Conf. Artif. Intell, vol. 38, 2024, pp. 1417–1425.
[47] X. Huang, X. Wang, Q. Teng, X. He, H. Chen, Degradation type-aware image restoration for effective object detection in adverse weather, Sensors 24 (19) (2024) 6330.
[48] A. Wu, C. Deng, Single-domain generalized object detection in urban scene via cyclic-disentangled self-distillation, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2022, pp. 847–856.
[49] Y. Liu, S. Zhou, X. Liu, C. Hao, B. Fan, J. Tian, Unbiased faster r-cnn for single-source domain generalized object detection, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2024, pp. 28838–28847.
[50] J. Kim, E. Cho, S. Kim, H.J. Kim, Retrieval-augmented open-vocabulary object detection, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2024, pp. 17427–17436.
[51] M. Zhang, R. Liu, D. Deguchi, H. Murase, Texture-guided transfer learning for low-quality face recognition, IEEE Trans. Image Process. 33 (2023) 95–107.
[52] Z. Yang, Z. Li, X. Jiang, Y. Gong, Z. Yuan, D. Zhao, C. Yuan, Focal and global knowledge distillation for detectors, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2022, pp. 4643–4652.
[53] Z. Zhang, S. Mei, M. Ma, Z. Han, Adaptive composite feature generation for object detection in remote sensing images, IEEE Trans. Geosci. Remote Sens. 62 (2024).
[54] Z. Liu, Y. Wang, X. Chu, N. Dong, S. Qi, H. Ling, A simple and generic framework for feature distillation via channel-wise transformation, in: Proc. IEEE/CVF Int. Conf. Comput. Vis, 2023, pp. 1129–1138.
[55] L. Zhang, K. Ma, Improve object detection with feature-based knowledge distillation: towards accurate and efficient detectors, in: Proc. Int. Conf. Learn. Represent, 2020.
[56] M. Hong, Y. Xie, C. Li, Y. Qu, Distilling image dehazing with heterogeneous task imitation, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2020, pp. 3462–3471.
[57] W. Lee, J. Lee, D. Kim, B. Ham, Learning with privileged information for efficient image super-resolution, in: Proc. Eur. Conf. Comput. Vis, Springer, 2020, pp. 465–482.
[58] J. Zhou, B. Zhang, D. Zhang, G. Vivone, Q. Jiang, Dtkd-net: dual-teacher knowledge distillation lightweight network for water-related Optics image enhancement, IEEE Trans. Geosci. Remote Sens. 62 (2024).
[59] Z. Chen, Y. Qian, X. Yang, C. Wang, M. Yang, Amfd: Distillation via adaptive multimodal fusion for multispectral pedestrian detection, arXiv preprint arXiv:2405.12944, 2024.
[60] X. Liu, J. Wang, B. Leng, S. Zhang, Dual-modeling decouple distillation for unsupervised anomaly detection, in: Proc. ACM Int. Conf. Multimedia, 2024, pp. 5035–5044.
[61] C.-Y. Wang, H.-Y.M. Liao, Y.-H. Wu, P.-Y. Chen, J.-W. Hsieh, I.-H. Yeh, Cspnet: a new backbone that can enhance learning capability of CNN, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Worksh, 2020, pp. 390–391.
[62] Z. Ge, S. Liu, F. Wang, Z. Li, J. Sun, Yolox: Exceeding yolo series in 2021, arXiv preprint arXiv:2107.08430, 2021.
[63] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proc. IEEE Int. Conf. Comput. Vis, 2017, pp. 2980–2988.
[64] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, Masked autoencoders are scalable vision learners, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2022, pp. 16000–16009.
[65] R. Quan, X. Yu, Y. Liang, Y. Yang, Removing raindrops and rain streaks in one go, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit, 2021, pp. 9147–9156.
[66] S.-C. Huang, T.-H. Le, D.-W. Jaw, Dsnet: joint semantic learning for object detection in inclement weather conditions, IEEE Trans. Pattern Anal. Mach. Intell. 43 (8) (2020) 2623–2633.
[67] W.-T. Chen, H.-Y. Fang, C.-L. Hsieh, C.-C. Tsai, I. Chen, J.-J. Ding, S.-Y. Kuo, et al., All snow removed: single image desnowing algorithm using hierarchical dual-tree complex wavelet representation and contradict channel loss, in: Proc. IEEE/CVF Int. Conf. Comput. Vis, 2021, pp. 4196–4205.
[68] M. Everingham, L. Van Gool, C.K.I. Williams, J. Winn, A. Zisserman, The pascal visual object classes (VOC) challenge, Int. J. Comput. Vis. 88 (2010) 303–338.
[69] B. Li, W. Ren, D. Fu, D. Tao, D. Feng, W. Zeng, Z. Wang, Benchmarking single-image dehazing and beyond, IEEE Trans. Image Process. 28 (1) (2018) 492–505.

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